

Insect Colour Selection Shaped by Environmental Floral Hue Ratios.pdf

by Yuki Wu

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Coursework Coversheet

Name	Xiaoqi (Yuki) Wu
Date	31/10/2024
Module	African Biology Field Course
Coursework item	Mini-project

Self-reflection is a non-assessed part of the coursework submission to aid your learning

1. **Have you used previous feedback/self-reflection to produce this coursework? If so, how? If not, why?**

Yes, I utilized my previous feedback to improve my time management by creating a well-organized schedule for completing this project.

2. **What do you feel went well?** (e.g., development of specific skills, sense of achievement, part of the assignment you are most proud of)

I did well with time management. I made a point to complete one section every day, allowing ample time for essay revisions.

3. **What do you feel did not go well?** (e.g., learning outcomes that you found most difficult to meet, aspects of the assignment that were challenging, time-management)

What I found challenging was finding related papers to discuss, as there were limited papers had conducted experiments to investigate the insect preference in colour ratio. Therefore, I discussed more on the limitations of our experiment. Additionally, I discovered that it is quite difficult to make an experiment perfect, given the multitude of variables that need to be considered and controlled.

4. **What did you hope to learn from doing this coursework? Did you achieve that?**

I anticipated learning about the considerations necessary for experiment design. Although designing and completing the experiment was challenging under limited time and experimental equipment, we still successfully completed our coursework and I really enjoy investigating insects with my teammates.

Insect Colour Selection Shaped by Environmental Floral Hue Ratios

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INTRODUCTION

Insects play a critical role as pollinators, facilitating the reproductive success of flowering plants by transferring pollen. This pollination process often relies on visual cues, with many insects exhibiting preferences for specific flower colours (Papiorek et al., 2016; Reverté et al., 2016). These colour preferences are fundamental in shaping plant-insect interactions, which maintain ecosystem biodiversity. However, pollinator populations are currently declining (Potts et al., 2010), posing a threat to both wild and agricultural ecosystems. Therefore, identifying the factors that influence pollinator behaviour is essential for developing targeted conservation strategies to support both wild and agricultural ecosystems.

In this study, we hypothesize that insect attraction to flowers aligns with the prevalent flower colours in their environment, an adaptation that may have evolved to enhance foraging efficiency. Specifically, we propose that insects use these dominant colours as indicators for food sources, optimizing their energy intake per unit of time (Chittka & Menzel, 1992). If our hypothesis holds, insect attraction should correlate with environmental colour ratios, suggesting that insects have adapted to focus on the most abundant floral resources to enhance foraging efficiency.

Our study tests this hypothesis at the Klipbakkop site by simulating flowers of different hues using coloured pan traps, examining the correlation between insect attraction and the colour ratios in the surrounding flora. A positive correlation between colour ratios in the environment and insect abundance in corresponding-coloured traps would support the idea that insects use environmental colour cues in their foraging decisions, highlighting a widespread and generalizable mechanism in pollinator behaviour.

Findings from this study aim to deepen our understanding of insect colour preferences in natural foraging and contribute insights into pollinator behaviour and conservation strategies for diverse ecosystems.

METHODS

Experiment Procedure and Data Collection

Data were collected over three days, beginning daily at 9:00 AM. Each day, we randomly selected one burnt and one unburnt site, which we expected to have different floral colour ratios. A 75 m transect was set up at each site with random starting points and directions to mitigate spatial autocorrelation bias and better reflect the variability within each habitat.

To quantify colour abundance, we used a standardised colour unit system: plants with flowers under 10 cm or only with one flower were recorded as one unit, those between 10-20 cm as two units, and so forth. Flower colours (red, yellow, blue, white) were recorded as units based on plant size, providing a quantitative measure of colour ratios across transects.

At every 15-metre point along each transect, GPS coordinates were recorded, and a circular 5 m radius sampling zone was established (Fig.1). Within these zones, team members individually documented colour units in designated sections to avoid overlapping (Fig. 2). After each 15-metre point, we summed colour units, ensuring circular sampling zones of five metres in radius, spaced five metres apart to minimise overlap. This sampling design minimized redundancy and ensured comprehensive coverage of each site.

We deployed pan traps in five colours (red, yellow, blue, white, and green), with green serving as a control to gauge accidental insect entry. Preliminary testing supported green's effectiveness as a control for background rates. After recording colour units at each point, we randomly placed the pan traps around the centre (Fig. 2). The traps were filled with a 3 cm depth of diluted detergent when all colour units were counted along the transect to standardise collection time.

The traps were left until 4:00 PM, and captured insects were stored in labelled tubes. Insects were identified to the order level, and terrestrial insects, considered accidental entries, were excluded from the analysis. This left only aerial insects for analysis, which have a broader field of view and are more likely to actively select pan traps.



Figure 1. Map of sample points around the Klipbökkop reserve. The map displays the 36 sample

points distributed across six different sites, where transects were established and six sets of data were collected per transect. Red pins represent sample points in burnt areas, while green pins indicate unburnt areas.

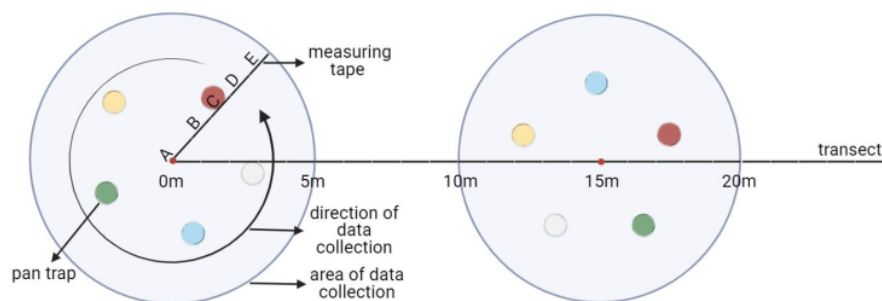


Figure 2. Layout of the transect and data collection setup. The diagram illustrates the method used at the 0 m point along the transect. Five team members (A, B, C, D, and E) collected data within a 5 m radius around the origin, each covering a 1 m section of the measuring tape and moving in a circular path to record colour units. Team member A stood at the centre (origin). After recording, pan traps were randomly placed within the collection area.

Statistical Analysis

To assess the impact of floral colour ratios on insect attraction, we calculated insect abundance percentages for each colour pan trap, using a green trap as a control to estimate baseline accidental entries. To isolate true colour attraction, we subtracted the green baseline (8.94%) from each colour pan's abundance percentage.

Using these adjusted values, we conducted two statistical analyses: one-sample t-tests compared each adjusted colour abundance to the green baseline, identifying if each colour attracted insects above the background rate. Pearson correlation analyses examined if environmental floral colour ratios were associated with increased insect attraction to matching pan colours.

Ethical Note

All insect specimens were ethically sourced through pan trapping, an established method for studying insect diversity with minimal environmental impact. No regulatory approval was required. Traps were set temporarily and monitored to prevent capturing endangered or non-target species. Insects were handled responsibly, and efforts were made to limit disruption to local populations and habitats. Trap duration was kept short, adhering to local biodiversity guidelines to ensure minimal disturbance.

RESULTS

Background Rate and Adjusted Insect Abundance

To account for accidental insect entries, the mean percentage of insect abundance in the green (control) pan trap, which reflects the background rate, was calculated as 8.94%. This rate was subtracted from each colour pan trap (red, yellow, blue, and white) to yield adjusted insect abundance percentages for each colour (Table 1).

Table 1: Background-Adjusted Mean Insect Abundance Percentages by Pan Trap Colour

Pan Trap Colour	Mean Insect Abundance (%)	Adjusted Mean Insect Abundance (%)
Green (Control)	8.94	-
Red	8.97	0.03
Yellow	29.11	20.17
Blue	33.00	24.06
White	19.98	11.04

One-Sample t-Test Analysis

One sample t-tests were performed to determine whether the insect abundance in each colour pan trap (red, yellow, blue, and white) was significantly higher than the background rate (green pan trap). The results indicate significant differences for yellow ($t = 8.163$, $p < 0.001$), blue ($t = 8.647$, $p < 0.001$), and white ($t = 4.113$, $p < 0.001$), meaning that insect abundance in these colour traps was significantly higher than the background rate. In contrast, the t-test for red ($t = 0.022$, $p = 0.982$) did not show a significant difference, suggesting that the insect abundance in the red pan trap was similar to the background rate (Table 2).

Table 2: One-Sample t-Test Results for Each Colour Pan Trap Compared to Background Rate

Pan Trap Colour	t-Statistic	p-Value	Interpretation
Red	0.022	0.982	Not significantly different from background
Yellow	8.163	< 0.001	Significantly higher than background
Blue	8.647	< 0.001	Significantly higher than background
White	4.113	< 0.001	Significantly higher than background

Correlation Analysis

Pearson correlation analyses were conducted between the environmental colour ratios and the adjusted insect abundance for each colour. The results indicate a positive and significant correlation only for yellow ($r = 0.369$, $p = 0.027$), suggesting that a higher percentage of yellow flowers in the environment is associated with increased insect abundance in the yellow pan trap. In contrast, the correlations for blue ($r = -0.103$, $p = 0.551$), white ($r = 0.034$, $p = 0.843$), and red ($r = 0.222$, $p = 0.194$) were not significant, indicating no statistically significant relationship between these environmental flower colour ratios and insect abundance (Table 3, Fig. 3).

Table 3: Correlation Analysis Results for Environmental Colour Ratios and Adjusted Insect Abundance

Pan Trap Colour	Adjusted Correlation Coefficient (r)	p-Value	Interpretation
Red	-0.103	0.551	Not significant
Yellow	0.369	0.027	Positive, significant ($p < 0.05$)
Blue	0.034	0.843	Not significant
White	0.222	0.194	Not significant

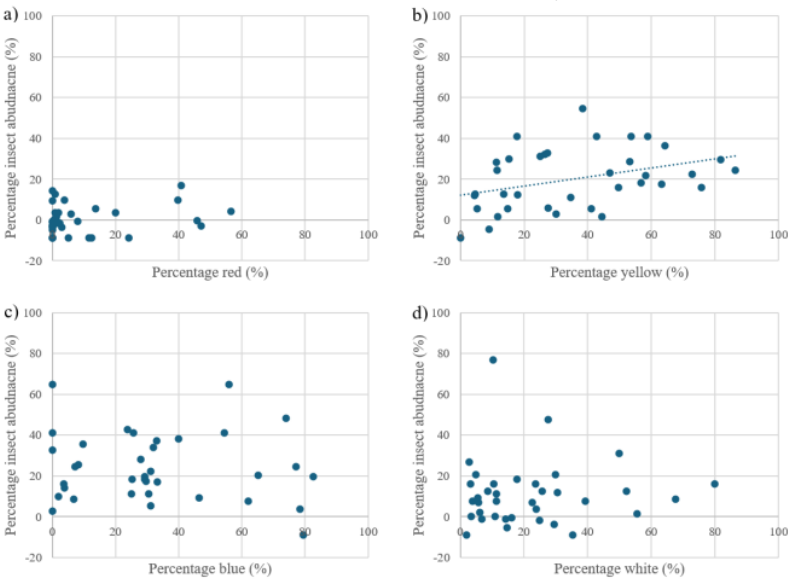


Figure 3. Scatter plots illustrating the relationship between environmental flower colour percentages and corresponding insect abundance percentages in pan traps. Each panel represents a different flower colour: **(a)** red, **(b)** yellow, **(c)** blue, and **(d)** white. The y-axis indicates the percentage of insect abundance in each coloured pan trap, while the x-axis shows the percentage of that flower colour in the environment. Only panel (b) for yellow reveals a slight positive trend, suggesting a potential association between yellow flower abundance and insect attraction to yellow.

DISCUSSION

This study explored how environmental flower colour ratios influence insect attraction using coloured pan traps. Red pan traps showed no significant difference from the background rate, likely because many insects, especially bees, are less sensitive to red, making it reasonable to expect no significant correlation with insect abundance (Chittka & Waser, 1997).

Blue and white pan traps showed significant differences from the background rate but no correlation with environmental colour ratios, suggesting that insect attraction to these colours may not depend on local abundance. This could indicate that many insects, especially generalist foragers, find blue and white universally appealing due to high visibility or perceived nectar access, reflecting fixed colour preferences rather than adjustments to local floral conditions (Garcia et al., 2022).

In contrast, the significant positive correlation for yellow suggests that insects may be particularly responsive to yellow, potentially due to their association with high nectar rewards and visibility under natural light. This preference may reflect an adaptive foraging strategy, especially for pollinators such as bumble bees that are responsive to yellow wavelengths (Vickery & Vickery, 1992).

Several limitations may have influenced the results. First, the artificial nature of pan traps differs from real flowers in aspects like scent, shape, and height, which could impact insect attraction. Using odourless detergent controlled for scent bias; however, natural floral scents are used alongside colour to locate flowers, and this study's focus on colour preference might not fully capture natural foraging behaviour (Hirota et al., 2012). Future studies could incorporate artificial floral scents to better simulate real flowers.

Additionally, pan trap sampling may have introduced spatial autocorrelation, as insects likely encountered multiple traps within their foraging range. Randomizing transect placement and site selection was intended to reduce this bias; however, increasing the distance between traps or implementing more advanced randomization techniques, such as stratified random sampling, could further minimize spatial autocorrelation.

Our sample size was small, limiting statistical power, and combining burnt and unburnt sites may have introduced variability, as these environments differ in insect and plant diversity (Lazarina et al., 2017). Sampling time also varied, with the third day starting an hour earlier, possibly affecting results due to insect diel activity patterns, where insect activity fluctuates at different times of day (Beck, 2012). Additionally, weather conditions such as wind or cloud cover, which can influence insect activity, were not controlled, and colour saturation or reflectance differences between pan traps and real flowers could have influenced insect responses (McCall & Primack, 1992; Papiorek et al., 2016).

Overall, this study highlights the complexity of insect colour preferences and suggests that while some colours like yellow align with environmental cues, others like blue and white may represent broader, less environmentally specific attraction patterns. Further studies incorporating more naturalistic floral cues, such as scent and structural elements, would help clarify how these colour preferences translate to natural foraging behaviour and enhance our understanding of insect-plant interactions in diverse ecosystems.

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GRADEMARK REPORT

FINAL GRADE

GENERAL COMMENTS

65/100

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GRADING FORM: LIFESCIGRADINGFORMMINORV1

YUKI WU

CRITERIA MET

How does this work meet the criteria for the grade it has been given?



An abstract would be good. Good formulation of hypotheses. Good fig. 2. No need to ethical note for these insects. Mainly accurate and well organised. Inconsistencies in reference section.

AREAS TO IMPROVE

How could this work – and similar work in future – be improved, to gain a higher mark?



add more legend to figure